PSPM 2016/17

Q1. $$\frac{(p+qi)}{3i} = (3+\sqrt{-16}-i^3)$$

$$p+qi = (3+\sqrt{-16}-i^3)(3i)$$ K1

$$p+qi = (3+\sqrt{16}\sqrt{-1}-i^3)(3i)$$

$$p+qi = (3+4i-i^3)(3i)$$ K1

$$p+qi = 9i+12i^2-3i^4$$

$$p+qi = 9i+12(-1)-3(1)$$ K1

$$p+qi = 9i-15$$ K1

$$p+qi = -15+9i$$

$$\therefore p=-15, \ q=9$$ J1 J1

6M

Q2. $$3^{2x-1} = 4(3^x)-9$$

$$3^{2x}\cdot 3^{-1} = 4(3^x)-9$$ K1

$$(3^x)^2(3^{-1}) = 4(3^x)-9$$

Let $u=3^x$

$$\frac{1}{3}(u)^2 = 4u-9$$ K1

$$\frac{1}{3}u^2-4u+9=0$$

$$u^2-12u+27=0$$ K1

$$(u-3)(u-9)=0$$

$$u=3, \quad u=9$$ K1

$$3^x=3^1 \qquad 3^x=9$$

$$3^x=3^2$$

$$\therefore x=1 \qquad x=2$$ J1 J1

6M

Q3. (a) $ff^{-1}(x) = x$

$$\frac{f^{-1}(x)^2+1}{5} = x \qquad K1$$

$$f^{-1}(x)^2+1 = 5x$$

$$f^{-1}(x)^2 = 5x-1 \qquad K1$$

$$\therefore\ f^{-1}(x) = \sqrt{5x-1} \qquad J1$$

$$fg(x) = \frac{1}{5}(e^{2(3x-1)}+1)$$

$$\frac{g(x)^2+1}{5} = \frac{1}{5}(e^{2(3x-1)}+1) \qquad K1$$

$$g(x)^2+1 = e^{2(3x-1)}+1$$

$$g(x)^2 = e^{2(3x-1)}$$

$$g(x) = e^{3x-1} \qquad J1 \qquad 5M$$

(b) $$gf(2) = g\left[\frac{2^2+1}{5}\right] = g(1) \qquad K1$$

$$= e^{3(1)-1} = e^2 \qquad K1,\ J1 \qquad 3M$$

(c) $gg^{-1}(x) = x$

$$e^{3g^{-1}(x)+1} = x$$

$$3g^{-1}(x)+1 = \ln x \qquad K1$$

$$g^{-1}(x) = \frac{\ln x - 1}{3} \qquad J1$$

since $x \geqslant 0$,

$$D_{g^{-1}(x)} = [\tfrac{1}{e}, \infty) \quad R_{g^{-1}(x)} = [0, \infty) \qquad J1 \qquad 3M$$

Q4. (a) $\sqrt{7-3\sqrt{5}} = \sqrt{x} - \sqrt{y}$

$$\left(\sqrt{7-3\sqrt{5}}\right)^2 = (\sqrt{x} - \sqrt{y})^2$$ K1

$$7-3\sqrt{5} = x - 2\sqrt{x}\sqrt{y} + y$$

$$7-3\sqrt{5} = x+y-2\sqrt{xy}$$

<u>compare:</u>

$7 = x+y$

$y = 7-x \quad — ①$

$(-3\sqrt{5})^2 = (-2\sqrt{xy})^2$ K1

$9(5) = 4(xy)$

$4xy = 45 \quad — ②$

<u>SUBS ① INTO ②</u>

$$4x(7-x) = 45$$

$$-4x^2+28x = 45$$

$$4x^2-28x-45=0$$ K1

$$(2x-9)(2x-5)=0$$

$$x = \frac{9}{2}, \quad x = \frac{5}{2}$$ J1

$$y = 7-\frac{9}{2}, \quad y = 7-\frac{5}{2}$$ K1

$$y = \frac{5}{2}, \quad y = \frac{9}{2}$$

$\therefore$ $x = \frac{9}{2}, y = \frac{5}{2}$ or $x = \frac{5}{2}, y = \frac{9}{2}$ J1

6M

Q4. (b) $\log_2 x - \log_4(3x+4) = 0$

$$\log_2 x - \frac{\log_2(3x+4)}{\log_2 4} = 0 \qquad \text{K1}$$

$$\log_2 x - \frac{\log_2(3x+4)}{2\log_2 2} = 0 \qquad \text{K1}$$

$$2\log_2 x - \log_2(3x+4) = 0$$

$$\log_2 x^2 = \log_2(3x+4) \qquad \text{K1}$$

$$x^2 = (3x+4) \qquad \text{K1}$$

$$x^2 - 3x - 4 = 0 \qquad \text{K1}$$

$$(x-4)(x+1) = 0$$

$$x = 4, \quad x = -1 \qquad \text{K1}$$

$$\therefore \underline{\underline{x = 4}} \qquad \text{J1} \qquad \text{7M}$$

Q5. (a) $\left|\frac{3}{x-4}\right| = 7, \; x \neq 4.$

$$\frac{3}{x-4} = 7 \quad \text{or} \quad \frac{3}{x-4} = -7 \qquad \text{B1}$$

$$3 = 7x - 28 \qquad\qquad 3 = -7x + 28 \qquad \text{K1}$$

$$7x = 31 \qquad\qquad 7x = 25 \qquad \text{K1}$$

$$\underline{\underline{x = \frac{31}{7}}} \qquad\qquad \underline{\underline{x = \frac{25}{7}}} \qquad \text{J1, J1} \qquad \text{5M}$$

Q5 (b) $\dfrac{-4-x}{x-3} \geq x+4, \quad x \neq 3$

$$\frac{-4-x}{x-3} - (x+4) \geq 0 \qquad \text{K1}$$

$$\frac{-4-x-(x+4)(x-3)}{x-3} \geq 0 \qquad \text{K1}$$

$$\frac{-4-x-(x^2+x-12)}{x-3} \geq 0$$

$$\frac{-x^2-2x+8}{x-3} \geq 0 \qquad \text{K1}$$

$$\frac{x^2+2x-8}{x-3} \leq 0 \qquad \text{K1}$$

$$\frac{(x+4)(x-2)}{x-3} \leq 0 \qquad \text{K1}$$

$x=-4, \quad x=2, \quad x \neq 3$

	$(-\infty,-4]$	$[-4,2]$	$[2,3)$	$(3,\infty)$
	-5	-3	$2\cdot5$	4
$x+4$	$-$	$+$	$+$	$+$
$x-2$	$-$	$-$	$+$	$+$
$x-3$	$-$	$-$	$-$	$+$
	$-$	$+$	$-$	$+$

K1 J1

$\therefore$ Solution set:

$$\{x : x \leq -4 \text{ or } 2 \leq x < 3\} \qquad \text{J1}$$

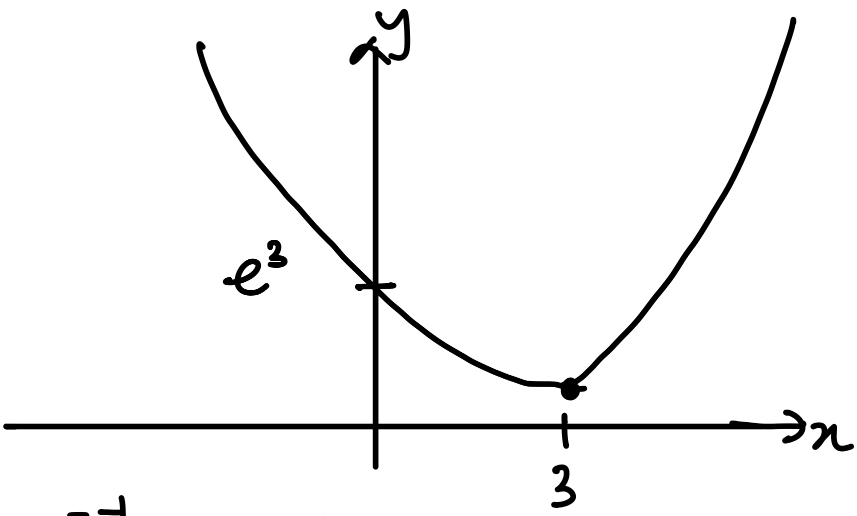
8m

Q6. (a) $f(x) = e^x$, $g(x) = |x-3|$

$$|x-3| = \begin{cases} x-3, & x \geq 3 \\ -(x-3), & x < 3 \end{cases}$$ K1

$$f \circ g(x) = e^{g(x)}$$
$$= e^{|x-3|}$$
$$= \begin{cases} e^{x-3}, & x \geq 3 \\ e^{-(x-3)}, & x < 3 \end{cases}$$ J1 2m

(b)



R1

R1

$[f \circ g(x)]^{-1}$ exist for : $[3, \infty)$ or $(-\infty, 3]$ J1 3m

(c) For $x \geq 3$,

$$f \circ g(x) = e^{x-3}$$
let $f \circ g(x) = h(x)$
$$hh^{-1}(x) = x$$ K1
$$e^{h^{-1}(x)-3} = x$$
$$h^{-1}(x) - 3 = \ln x$$ K1
$$h^{-1}(x) = \ln x + 3$$
$$\therefore [f \circ g(x)]^{-1} = \ln(x) + 3$$ J1 3m

Q6(a) $h \circ f(x) = \dfrac{2e^x}{1-3e^x}$

$$h[e^x] = \frac{2e^x}{1-3e^x}$$

Let, $u = e^x$ K1

$x = \ln u$

$$h[u] = \frac{2e^{\ln u}}{1-3e^{\ln u}}$$ K1

$$h[u] = \frac{2u}{1-3u}$$

$$\therefore \quad h[x] = \frac{2x}{1-3x}$$ J1

Let $h(x_1) = h(x_2)$

$$\frac{2x_1}{1-3x_1} = \frac{2x_2}{1-3x_2}$$

$$2(x_1)(1-3x_2) = 2x_2(1-3x_1)$$ K1

$$x_1 - 3x_1x_2 = x_2 - 3x_1x_2$$

$$x_1 = x_2 - 3x_1x_2 + 3x_1x_2$$

$$\therefore \quad x_1 = x_2$$ J1

Since $x_1 = x_2$, $h(x)$ is a one to one function. 5M

Q7.(a) $\lim_{x \to 2} \frac{x-2}{x^4-16} = \lim_{x \to 2} \frac{x-2}{(x^2+4)(x^2-4)}$

$= \lim_{x \to 2} \frac{x-2}{(x^2+4)(x+2)(x-2)}$ K1

$= \lim_{x \to 2} \frac{1}{(x^2+4)(x+2)}$

$= \frac{1}{(2^2+4)(2+2)}$ K1

$= \underline{\frac{1}{32}}$ J1 3M

Q7.(b) $\lim_{x \to \infty} \frac{(2-x)(x-1)}{(x-3)^2} = \lim_{x \to \infty} \frac{2x-2-x^2+x}{x^2-6x+9}$

$= \lim_{x \to \infty} \frac{-x^2+3x-2}{x^2-6x+9}$

$= \lim_{x \to \infty} \frac{\frac{-x^2+3x-2}{x^2}}{\frac{x^2-6x+9}{x^2}}$ K1

$= \lim_{x \to \infty} \frac{-1+\frac{3}{x}-\frac{2}{x^2}}{1-\frac{6}{x}+\frac{9}{x^2}}$

$= \frac{-1+0-0}{1-0+0}$ K1

$\underline{= -1}$ J1 3M

Q8. (a) $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} \frac{1}{2-\sqrt{x}}$ K1

$= 0$ K1

$\therefore$ $y=0$ is the horizontal asymptote J1 3m

Q8. (b) $f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{1}{2-\sqrt{x+h}} - \frac{1}{2-\sqrt{x}}\right]$$

$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{(2-\sqrt{x})-(2-\sqrt{x+h})}{(2-\sqrt{x+h})(2-\sqrt{x})}\right]$$

$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{-\sqrt{x}+\sqrt{x+h}}{(2-\sqrt{x+h})(2-\sqrt{x})} \times \frac{\sqrt{x+h}+\sqrt{x}}{\sqrt{x+h}+\sqrt{x}}\right]$$ K1

$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{-\sqrt{x}\sqrt{x+h}-(x)+(x+h)+\sqrt{x}\sqrt{x+h}}{(2-\sqrt{x+h})(2-\sqrt{x})(\sqrt{x+h}+\sqrt{x})}\right]$$

$$= \lim_{h\to 0} \frac{1}{h}\left[\frac{h}{(2-\sqrt{x+h})(2-\sqrt{x})(\sqrt{x+h}+\sqrt{x})}\right]$$ K1

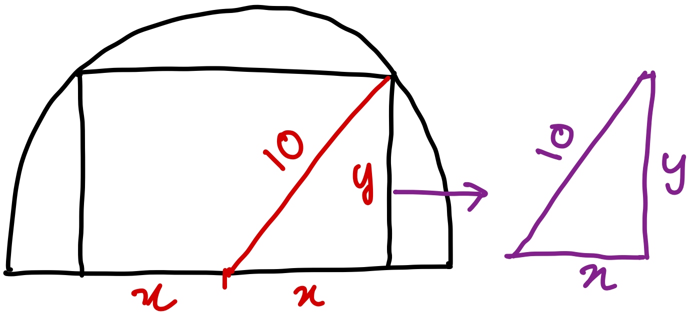
$$= \lim_{h\to 0} \frac{1}{(2-\sqrt{x+h})(2-\sqrt{x})(\sqrt{x+h}+\sqrt{x})}$$

$$= \frac{1}{(2-\sqrt{x})(2-\sqrt{x})(\sqrt{x}+\sqrt{x})}$$ K1

$$= \frac{1}{2\sqrt{x}(2-\sqrt{x})^2}$$ J1

4m

Q9(a)



$$x^2+y^2=100$$

$$y^2=100-x^2$$

$$y=(100-x^2)^{\frac{1}{2}}$$

$$A=2xy$$

$$A=2x(100-x^2)^{\frac{1}{2}}$$

$u=2x$ $\quad$ $v=(100-x^2)^{\frac{1}{2}}$

$u'=2$ $\quad$ $v'=\frac{1}{2}(100-x^2)^{-\frac{1}{2}}(-2x)$

$$=-\frac{x}{\sqrt{100-x^2}}$$

① $\frac{dA}{dx}=\sqrt{100-x^2}(2)+(2x)\left(-\frac{x}{\sqrt{100-x^2}}\right)$ K1

$$=2\sqrt{100-x^2}-\frac{2x^2}{\sqrt{100-x^2}}$$

② $\frac{dA}{dx}=0$

$$2\sqrt{100-x^2}-\frac{2x^2}{\sqrt{100-x^2}}=0$$

$$2\sqrt{100-x^2}=\frac{2x^2}{\sqrt{100-x^2}}$$

$$2(100-x^2) = 2x^2$$
$$200-2x^2 = 2x^2$$
$$4x^2-200 = 0$$
$$x^2 = 50$$
$$x = \pm\sqrt{50},\ x = \sqrt{50}$$ K1

③ $$\frac{dA}{dx} = 2\sqrt{100-x^2} - \frac{2x^2}{\sqrt{100-x^2}}$$ K1

$$\frac{dA}{dx} = \frac{2(100-x^2)-2x^2}{\sqrt{100-x^2}}$$

$$\frac{dA}{dx} = \frac{200-4x^2}{\sqrt{100-x^2}} \begin{matrix} \leftarrow u \\ \leftarrow v \end{matrix}$$

$$u = 200-4x^2 \qquad v = (100-x^2)^{\frac{1}{2}}$$
$$u' = -8x \qquad v' = \frac{1}{2}(100-x^2)^{-\frac{1}{2}}(-2x)$$
$$= -\frac{x}{\sqrt{100-x^2}}$$

$$\frac{d^2A}{dx^2} = \frac{\sqrt{100-x^2}(-8x) - (200-4x^2)\left[-\frac{x}{\sqrt{100-x^2}}\right]}{100-x^2}$$

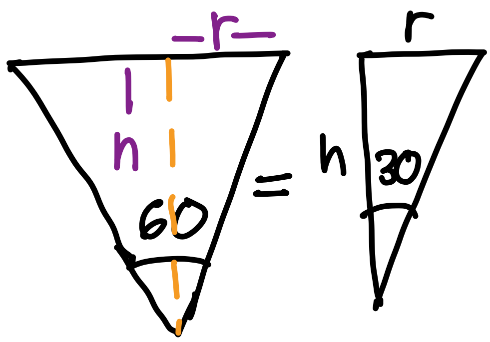
$$\left.\frac{d^2A}{dx^2}\right|_{x=\sqrt{50}} = \frac{-400-0}{50} = -8 < 0 \text{ (max)}$$ K1

$$A = 2x(100-x^2)^{\frac{1}{2}}$$
$$= 2\sqrt{50}(100-50)^{\frac{1}{2}}$$
$$= 100 \text{ cm}^2$$ J1

5m

Q9(b) $\frac{dV}{dt} = -6$, $\frac{dh}{dt} = ?$ when $V = 24\pi$

$$\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$$



$$\frac{r}{h} = \tan 30$$

$$r = \frac{h}{\sqrt{3}}$$ K1

$$V = \frac{1}{3}\pi r^2 h$$

$$V = \frac{1}{3}\pi\left(\frac{h}{\sqrt{3}}\right)^2 h$$ K1

$$= \frac{1}{3}\pi\left(\frac{h^3}{3}\right)$$

$$V = \frac{1}{9}\pi h^3 \Rightarrow \frac{dV}{dh} = \frac{1}{3}\pi h^2$$ K1

$$\frac{dh}{dV} = \frac{3}{\pi h^2}$$

When $V = 24\pi$

$$\frac{1}{9}\pi h^3 = 24\pi$$

$$\frac{h^3}{9} = 24 \Rightarrow h = 6$$ K1

$$\therefore \frac{dh}{dt} = \frac{3}{\pi h^2} \times -6$$ K1

$$= \frac{-18}{\pi h^2}$$

When $h = 6$

$$\frac{dh}{dt} = \frac{-18}{\pi(6^2)} = -\frac{1}{2\pi} \text{ cms}^{-1}$$ J1

6M

Q10(a) $x=e^{2t+1}$, $y=e^{-(2t-1)}$

$$\frac{dx}{dt}=2e^{2t+1} \qquad \frac{dy}{dt}=-2e^{-(2t-1)} \quad \text{K1}$$

$$\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}$$

$$=-2e^{-(2t-1)}\times\frac{1}{2e^{2t+1}} \quad \text{K1}$$

$$=\frac{-2e^{-(2t-1)}}{2e^{2t+1}}$$

$$=-\frac{1}{e^{4t}} \quad \text{J1}$$

when $t=1$

$$\frac{dy}{dx}=-\frac{1}{e^{4}} \quad \text{J1}$$

$$\frac{d^2y}{dx^2}=\frac{d}{dt}\left(-\frac{1}{e^{4t}}\right)\times\left(\frac{1}{2e^{2t+1}}\right)$$

$$=\frac{d}{dt}\left(-e^{-4t}\right)\left(\frac{1}{2e^{2t+1}}\right)$$

$$=4e^{-4t}\left(\frac{1}{2e^{2t+1}}\right) \quad \text{K1}$$

$$=\frac{4}{2e^{6t+1}}=\frac{2}{e^{6t+1}} \quad \text{J1}$$

when $t=1$

$$\frac{d^2y}{dx^2}=\frac{2}{e^{6(1)+1}}$$

$$=\frac{2}{e^{7}} \quad \text{J1}$$

7M

Q10(b) $z = x^2 - xy$

$= (e^{2t+1})^2 - (e^{2t+1})(e^{-(2t-1)})$ K1

$= e^{2(2t+1)} - e^{2t+1-2t+1}$

$= \underline{\underline{e^{4t+2} - e^2}}$ J1

$\frac{dz}{dt} = 4e^{4t+2}$ K1

For $\frac{dz}{dt} > 0$,

$4e^{4t+2} > 0$

$e^{4t+2} > 0$ K1

$\therefore \underline{\underline{\{t : t \in \mathbb{R}\}}}$ J1

5M

Q11 (a) $f(x) = \frac{2|x|}{x} + 5x$

$$|x| = \begin{cases} x & , x \geq 0 \\ -x & , x < 0 \end{cases}$$ B1

$$f(x) = \begin{cases} \frac{2x}{x} + 5x & , x \geq 0 \\ -\frac{2x}{x} + 5x & , x < 0 \end{cases}$$

$$= \begin{cases} 2 + 5x & , x \geq 0 \\ -2 + 5x & , x < 0 \end{cases}$$ K1

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0^+} 2 + 5x$$ K1

$$= 2 + 5(0)$$

$$= 2$$

$$\lim_{x \to 0^-} f(x) = \lim_{x \to 0^-} -2 + 5x$$ K1

$$= -2 + 5(0)$$

$$= -2$$

$\therefore$ Since $\lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$, $\lim_{x \to 0} f(x)$ does not exist, therefore f is not continuous at $x = 0$. J1

5m

Q11(b) $g(x) = \begin{cases} \sqrt{5-x}, & x<a \\ 3x-1, & x \geq a \end{cases}$

① $g(a) = 3(a)-1 = 3a-1$ K1

② $\lim\limits_{x \to a^+} g(x) = \lim\limits_{x \to a^+} 3x-1 = 3(a)-1 = 3a-1$ K1

$\lim\limits_{x \to a^-} g(x) = \lim\limits_{x \to a^-} \sqrt{5-x} = \sqrt{5-a}$ K1

since $g(x)$ is continuous at $x=a$,

$\lim\limits_{x \to a^-} g(x) = \lim\limits_{x \to a^+} g(x)$

$3a-1 = \sqrt{5-a}$

$(3a-1)^2 = (\sqrt{5-a})^2$ K1

$9a^2-6a+1 = 5-a$

$9a^2-5a-4=0$

$(9a+4)(a-1)=0$ K1

$a=-\frac{4}{9}, \ a=1$ J1

When $a=-\frac{4}{9}$, K1

$3(-\frac{4}{9})-1 \neq \sqrt{5+\frac{4}{9}}$

$\therefore a \neq -\frac{4}{9}$

When $a=1$,

$3(1)-1 = \sqrt{5-(1)}$

$\therefore \underline{\underline{a=1}}$ J1

8M

Q12.(a) $\tan x = \dfrac{\sin x}{\cos x}$

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$$

$u = \sin x \qquad v = \cos x$ K1

$u' = \cos x \qquad v' = -\sin x$

$$\frac{d}{dx}(\tan x) = \frac{\cos x(\cos x) - (\sin x)(-\sin x)}{\cos^2 x} \quad \text{K1}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x}$$

$$= \frac{1}{\cos^2 x} = \sec^2 x \quad \text{J1}$$

3M

(a) $y = \tan x$

$$\frac{dy}{dx} = \sec^2 x \quad \text{K1}$$

$$= 1 + \tan^2 x \quad \text{K1}$$

$$\frac{dy}{dx} = 1 + y^2 \quad \text{J1}$$

$$\therefore \frac{d^2y}{dx^2} = 2y\frac{dy}{dx} \quad \text{K1}$$

$$= 2y(1+y^2) \quad \text{J1}$$

For $\dfrac{d^2y}{dx^2} > 0$, $0 < x < \pi$

$2y(1+y^2) > 0$

since $1 + y^2 > 0$ K1

$2y > 0$

$y > 0$

$\tan x > 0$

$\therefore \left(0, \dfrac{\pi}{2}\right)$ J1

7M

Q12(b) $y = \tan(x+y)$

$$\frac{dy}{dx} = \sec^2(x+y)\frac{d}{dx}(x+y)$$ (k1)

$$= \sec^2(x+y)\left[1+\frac{dy}{dx}\right]$$

$$= \sec^2(x+y) + \sec^2(x+y)\frac{dy}{dx}$$

$$\frac{dy}{dx} - \sec^2(x+y)\frac{dy}{dx} = \sec^2(x+y)$$ (k1)

$$\frac{dy}{dx}\left[1-\sec^2(x+y)\right] = \sec^2(x+y)$$

$$\therefore \frac{dy}{dx} = \frac{\sec^2(x+y)}{1-\sec^2(x+y)}$$ (j1)

When $x = y = \alpha$,

$$\frac{dy}{dx} = \frac{\sec^2(\alpha+\alpha)}{1-\sec^2(\alpha+\alpha)}$$

$$= \frac{\sec^2(2\alpha)}{1-\sec^2(2\alpha)}$$

$$= \frac{\sec^2(2\alpha)}{-\tan^2(2\alpha)}$$ (k1)

Identities

$1+\tan^2\theta = \sec^2\theta$

$1-\sec^2\theta = \tan^2\theta$

$$= \frac{\left(\frac{1}{\cos^2 2\alpha}\right)}{-\left(\frac{\sin^2 2\alpha}{\cos^2 2\alpha}\right)}$$

$$= \frac{1}{\cos^2 2\alpha} \times \left(-\frac{\cos^2 2\alpha}{\sin^2 2\alpha}\right)$$

$$= -\frac{1}{\sin^2 2\alpha}$$

$$= \underline{\underline{-\operatorname{cosec}^2(2\alpha)}}$$ (j1)

5m